
How Fidelis Deception® Helps Defend Against AI-Accelerated Intrusions

Key Takeaways

- AI-assisted attacks are faster and harder to detect, using valid credentials and normal behavior to bypass traditional defenses.
- Fidelis Deception® flips detection logic by controlling what attackers see, turning reconnaissance into immediate detection.
- Any interaction with decoys triggers high-confidence alerts, eliminating false positives and reducing analyst fatigue.
- Deception disrupts every attack stage, from reconnaissance to lateral movement, without relying on signatures or baselines.
- The more thorough AI-driven attacks become, the more likely they are to expose themselves within the deception layer.

AI-powered attackers are faster and more systematic than ever. But they still trust what they see. Deception technology controls what they see.

87%

of security leaders say AI-related vulnerabilities grew faster than any other risk in 2025

44%

year-over-year rise in exploitation of public-facing applications in 2025

300K+

AI platform credentials exposed via infostealer malware on dark web in 2025

AI Attacks Are Getting Faster. Most Defenses Are Not Keeping Up.

Security teams who watch intrusion activity day in and day out will tell you something that does not make it into most reports: the most dangerous thing about modern attacks is not sophistication. It is speed combined with patience.

The World Economic Forum's Global Cybersecurity Outlook 2026, drawn from over 800 CISOs, CEOs, and security executives across 92 countries, found that 87% of respondents identified AI-related vulnerabilities as the fastest-growing cyber risk throughout 2025. 94% expect AI to be the single most significant force reshaping network security in the year ahead.^{[1](#)} The IBM X-Force Threat Intelligence Index 2026 explains why: exploitation of public-facing applications jumped 44% year-over-year, and AI tooling is actively compressing attacker decision cycles during reconnaissance, privilege escalation, and lateral movement.^{[2](#)}

This is not a theoretical risk. Most enterprise intrusion detection systems were not built for this threat landscape. Traditional security methods depend on recognizable patterns. Signature-based approaches need the threat to already exist in a database. Anomaly detection needs the

attacker to behave unusually. AI-assisted attackers are designed to avoid both: using valid credentials, native admin utilities, and legitimate SaaS integrations to blend into normal behavior while moving laterally through an environment.

Deception technology does not try to detect attacks by finding suspicious patterns. It controls what the attacker sees, reads, and trusts, turning their own reconnaissance process into the detection event.

AI Makes Attacks Faster. It Does Not Change What Attackers Still Need.

AI-assisted tooling makes specific intrusion phases faster and more precise. Reconnaissance is more systematic. Credential enumeration is more targeted. Lateral movement decisions, which path to take and which assets to prioritize, are increasingly guided by behavioral analysis on observed network data rather than slow manual trial and error.

This is documented in real campaigns. In July 2025, Ukraine's national CERT (CERT-UA) disclosed LAMEHUG, the first publicly documented malware to integrate a large language model directly into its attack flow.³ Attributed with moderate confidence to APT28 (Fancy Bear), LAMEHUG used an LLM to dynamically generate system reconnaissance commands, including hardware enumeration, process listing, and network connection mapping, without any hardcoded instructions. [Signature-based detection](#) and traditional intrusion prevention systems were structurally blind to it because there was no static pattern to match.³

What AI does not change is the dependency structure that every intrusion still runs on. Whether the attacker uses AI tooling or not, they need to map the environment, discover workable credentials, understand what assets are present, and find a viable path to their target. Every step requires the attacker to read and trust environmental signals: network topology, directory structures, service availability, file artifacts, credential stores.

Sophisticated attackers, even those augmented by AI agents, still trust the environment they see. That is exactly what [Fidelis Deception](#)® exploits.

See How Deception Technology Exposes Attackers Before They Reach Critical Assets

- Reconnaissance detection
- Deceptive credentials
- Lateral movement traps
- High-confidence alerts

[Download the Solution Brief](#)



SOLUTION BRIEF

Fidelis D[®]

Change the Gam

The best defense is a good technology gives you. Malware do have their place in the is different. It's a proactive environment and expose real-time, with minimal e control and reduces the deception technology is

Make Adversa

Fidelis Deception[®] site difficult and expensive an attacker's ability to convincing decoys and your organization gain

- Reliable alerts that
- Valuable time to un threat the attack.
- Critical intelligence continual improve
- Foundational cyber continuity, no ma



Fidelis Deception

Change the Game on Cyber Adversaries

How Fidelis Deception[®] Disrupts the Attack Lifecycle

- Attacker gains initial access Phishing or credential theft lands a low-privilege foothold. AI-assisted tools begin scanning and mapping the environment to plan lateral movement.
Initial access
- Reconnaissance hits deception layer Fidelis Deception[®] populates the network with decoy assets, fake AD accounts, and breadcrumbs that look identical to real infrastructure. The attacker's map is wrong from the start.
Recon exposure

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- Attacker pursues deceptive credentialsPoisoned breadcrumbs, fake memory credentials, and false AD entries appear as high-value targets during enumeration. The attacker interacts with them, believing they are real.
Credential deception
 - Alert fires the moment the decoy is touchedNo behavioral threshold. No baseline is required. Any interaction with a deceptive object generates a high-confidence alert. False positives are near zero by design.
High-confidence detection
 - Security team observes attacker TTPs in real timeWhile the attacker believes they are navigating real infrastructure, the security team tracks their moves, learns their techniques, and builds intelligence for hardening. Real assets stay untouched.
Threat intelligence

Why Signature-Based and Anomaly Detection Break Against AI Intrusions

AI is “lowering the barrier to entry, allowing less experienced groups to execute operations that once required advanced expertise,” and that adversaries will likely weaponize new capabilities faster than most enterprises can integrate defensive AI into mature, well-instrumented security programs.[2](#)

Signature-based methods

Intrusion detection systems that rely on signature-based methods need attackers to use known tools in known ways. [Living-off-the-land](#) intrusions that leverage native system utilities, and LLM-powered malware like LAMEHUG that generates commands dynamically rather than using hardcoded instructions, leave no signature to match. Zero-day attack detection is structurally limited here. The IBM X-Force 2026 report documents a surge in exploitation before public disclosure, with many vulnerabilities requiring no authentication at all to exploit.[2](#)

Anomaly detection

[Anomaly detection](#) and behavioral analytics are more theoretically sound, but operationally difficult. Enterprise behavioral baselines are noisy. AI-assisted attackers can study normal behavior in an environment and deliberately operate within it to avoid triggering thresholds. The practical result is high false positive rates that wear down analyst confidence over time and slow incident response exactly when speed matters most. Security teams end up chasing noise while real intrusion attempts progress undetected.

Deception asks a different question

Deception technology shifts the detection logic entirely. Instead of asking “does this network traffic look bad?” it asks “why is anything interacting with this object at all?” There is no legitimate reason for any real user or process to access a decoy asset, enumerate a fake Active Directory account, or attempt to authenticate with a poisoned credential. Any interaction with the deception layer is inherently malicious activity, regardless of whether it matches a known pattern or deviates from a baseline. That is how Fidelis Deception® dramatically [reduces false positives and improves detection accuracy](#) at the same time.

Traditional IDS / IPS

- Needs known signatures to detect threats
- High false positive rate in complex environments

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- Blind to living-off-the-land and zero-day attacks
 - Alert fatigue reduces detection accuracy over time
 - Reactive: requires a mistake in the real environment

Fidelis Deception®

- No signature required, detects by interaction logic
- Near-zero false positives by design
- Catches zero-day and AI-assisted attacks equally
- Every alert is a genuine call to action
- Proactive: creates the exposure point, does not wait for it

A Real Attack Scenario: What This Looks Like in Practice

Most intrusion scenarios feel abstract until you walk through one step by step. This is how a typical AI-assisted attack unfolds in an environment where Fidelis Deception is deployed.

Attack scenario with Fidelis Deception® deployed

Access A phishing payload lands on a workstation with standard domain credentials. Low-privilege foothold established. AI-assisted tools begin scanning network traffic and enumerating the environment. Recon The attacker's tools identify what appear to be high-value targets: a domain controller, shared drives, and AD accounts. Fidelis Deception® decoys are indistinguishable from the real assets alongside them. Credential grab Attacker picks up a deceptive credential from a fake AD account. Breadcrumb files and memory artifacts were placed specifically to attract this enumeration activity. Alert fires The moment that credential is used for lateral movement, Fidelis Deception® triggers a high-confidence alert. No threshold crossed. No baseline violated. The interaction itself is the signal. Response Security team knows the attacker's location, what they touched, and where they appear to be heading. Real assets remain untouched. The incident response team acts with full context, not guesswork.

What Fidelis Deception® Actually Covers Across Your Environment

The scenario above covers one path through one environment. In practice, Fidelis Deception creates exposure points across every stage an attacker depends on, from the first reconnaissance sweep to the final push toward sensitive data.

Reconnaissance and network mapping

Faster reconnaissance is one of the clearest advantages AI-assisted tooling provides. Systematic scanning, network topology mapping, and service enumeration all happen faster and more completely than manual methods. In a deception-rich environment, that thoroughness becomes a liability.

Fidelis Deception® maps the actual environment and uses terrain analysis informed by asset risk profiling to place decoys where attacker movement paths are most likely to intersect them. Decoys span laptops, servers, routers, cameras, printers, IoT devices, operating systems, applications, ports, and services across both on-premises and cloud environments. The more methodically an attacker scans, the more deceptive data they collect.

Credential discovery and Active Directory deception

IBM X-Force Threat Intelligence Index 2026 reported that infostealer malware exposed over 300,000 AI platform credentials on dark web markets in 2025 alone, reflecting how systematically attackers now pursue credential harvesting at scale.² After initial access, the move to credential discovery is nearly immediate. [Fidelis Deception® deploys fake Active Directory accounts](#), including Azure AD, alongside breadcrumbs designed to surface during enumeration: memory credentials, registry keys, documents, and file artifacts that appear as legitimate discovery targets. When those deceptive credentials are accessed or used, the activity generates a high-confidence alert. No interpretation required.

Lateral movement detection

Detecting lateral movement through conventional means is genuinely hard. Careful attackers who mimic legitimate administrative traffic can remain undetected for extended periods. AI-assisted tooling compounds this by optimizing movement paths to stay below behavioral thresholds. Deception degrades that optimization because the attacker has no way to know which assets are real and which are not. The more methodically they evaluate available paths, the more likely they are to interact with deceptive objects along the way. Fidelis Deception® [detects lateral movement](#) as it happens, not after the fact, and the alerts require no statistical interpretation.

Attacker TTP intelligence and ongoing improvement

Standard containment responses often destroy the forensic picture. Deception creates a different option: observe the attacker operating within the deception layer while real assets remain protected. Understanding which asset types an attacker prioritizes, which credential formats they pursue, and how they adapt when paths are blocked gives security teams intelligence for both the immediate incident response and longer-term hardening. Fidelis Deception® also supports Red Team and Blue Team risk simulations, allowing security teams to refine decoy placement and coverage over time based on what is actually observed in real engagements.

What This Means for Day-to-Day SOC Operations

Most advanced detection tools make analysts work harder before they work smarter. They need tuning, generate alert volume that must be sorted, and often require specialized expertise to operate. Against faster intrusion timelines, that overhead creates response lag that attackers can exploit.

Fidelis Deception® is designed to work the opposite way. The platform uses machine learning to analyze the environment, assess asset risk, and automate the deployment and ongoing updating of decoys and breadcrumbs based on that analysis. Security teams do not need to manually configure individual deceptive objects. Because any deception-layer alert reflects actual interaction with a decoy rather than a statistical anomaly, analysts can move directly to investigation without spending time validating whether the alert is real.

Telemetry, analysis, threat hunting, and incident response actions are consolidated in a single console. During an active event, every tool-switch introduces delay and the risk of dropped context. Threat hunting against deception-layer activity is backed by forensic visibility into attacker movements, which shortens investigation timelines and improves post-incident analysis quality.

For organizations running the [Fidelis Elevate XDR platform](#), deception-layer data correlates with network, endpoint, and sandbox signals, giving security teams cross-domain visibility across

cloud environments, on-premises infrastructure, and identity systems in a single operational picture.

Our customers detect post-breach attacks over 9x Faster

- Detect Advanced Threats Before Damage Escalates Trusted
- Cybersecurity Leader for 20+ Years
- See why security teams choose us over other solutions

[Request a Demo](#)[Read Datasheet](#)

The More Sophisticated the Attack, the Better Deception Works

There is a counterintuitive dynamic that does not get discussed enough. More sophisticated attackers, including those using AI-assisted tooling for systematic reconnaissance and thorough environment mapping, tend to interact with more of the environment, not less. They enumerate more. They evaluate more credential sources. They assess more paths before committing to lateral movement.

Every one of those interactions is a potential detection event in a deception-rich environment.

Defenders who rely entirely on catching attacker mistakes in the real environment are betting on adversary errors. Deception technology does not require errors. It creates structured exposure points that sit directly in the path of normal attacker methodology. The more thorough the attacker is, whether human-directed or AI-assisted, the more likely they are to surface in the detection layer.

Given where evolving cyber threats are heading, that structural advantage matters more than ever. Deception becomes more effective as attacks become more thorough, which is precisely the direction the threat data points.

1. [^https://www.weforum.org/publications/global-cybersecurity-outlook-2026/](https://www.weforum.org/publications/global-cybersecurity-outlook-2026/)
2. [^https://www.ibm.com/reports/threat-intelligence](https://www.ibm.com/reports/threat-intelligence)
3. [^https://thehackernews.com/2025/07/cert-ua-discovers-lamehug-malware.html](https://thehackernews.com/2025/07/cert-ua-discovers-lamehug-malware.html)